Aryan Agarwal

Undergraduate in Computer Science

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EDUCATION

Vellore Institute of Technology, Chennai
BTech in Computer Science
CGPA: 8.44/10 (Till 5th Sem) — Aug 2023 - Present

SKILLS

Programming Languages	Python, C++, C, Javascript
Backend	FastAPI, Express
Databases	MySQL, PostgreSQL Redis, MongoDB, Cassandra
Machine Learning	ML algorithms, Tensorflow and PyTorch, Data Preprocessing
DevOps	Docker, Nginx

EXPERIENCE

AI/ML Immersion Program — National University of Singapore (NUS) — *github.com/project*

Gained in depth knowledge about Neural Network architectures, their strengths, weaknesses and use cases. Further, different optimizers and loss functions were taught and explained for optimal selection of such functions during training and testing. Introduction to Tensorflow and Keras and implementation of Neural Networks in the respective libraries. For capstone project, a Deepfake Detection system was created by a team of 7 members. My responsibility in the project included training, validation, and performance evaluation of the ViT-Large-Patch16-224 vision transformer model within a deepfake detection system. The model's outputs were optimized for integration with an ensemble framework alongside an EfficientNet-B0 convolutional neural network. This work contributed to improved robustness, accuracy, precision, and recall of the overall deepfake detection pipeline.

PROJECTS

Microservice Architecture using Python — *github.com/application*

Architected a distributed ML application using Docker containers, separating UI and inference layers to mimic production-grade microservices architecture. Developed a REST-based communication layer between frontend and ML service containers over an isolated Docker network. Designed the system for scalability and extensibility, enabling independent scaling of the inference engine under increased user load. Applied cloud-native principles such as containerization, service abstraction, and stateless backend design.

Mini Jira – Full-Stack Project Management System — ***github.com/mini-jira***

Built **Mini Jira**, a full-stack project management system inspired by Jira, to streamline task tracking and team collaboration. Developed the backend using **Node.js**, **Express.js**, and **MongoDB**, following the **MVC architecture** for scalable and maintainable code organization. Integrated MongoDB through a **Docker container** to simplify deployment and environment consistency. Implemented secure authentication and authorization middleware using **JWT (JSON Web Tokens)**, enabling protected API access and session handling. Designed and tested RESTful APIs using **Postman** to ensure reliability and proper endpoint functionality. Additionally, implemented **pre-commit hooks** with **Unit Testing, Integration Testing, and Mutation Testing** to enforce code quality, improve test coverage, and maintain application stability throughout development.

CERTIFICATIONS

Google Cybersecurity Certificate - **(March 2025)**

Generative AI using IBM WatsonX – IBM (June 2025)

ACHIEVEMENTS AND EXTRACURRICULAR ACTIVITIES

- **National Semi-Finalist in 2025 Flipkart GRID 7.0** - Showcased problem solving skills in data structures and real-life problems at national level.
- **Core Member of Japanese Cultural Club – VIT Chennai -** Events and workshops towards Japan. Hosted Japanese exchange students on the campus.